CM5 handheld calculator carrier — top level

Stage status: Stage 1 research BLOCKED (official Raspberry Pi documents not accessible).
This project is a hierarchy skeleton only. Nothing here has been verified.
Notes: ../research/STAGE1_STATUS.md

CM5 Connectors

File: cm5_connectors.kicad_sch

Power

File: power.kicad_sch

Display DSI

File: display.kicad_sch

Camera CSI

File: camera.kicad_sch

Keypad

File: keypad.kicad_sch

USB Boot Control

File: usb_boot.kicad_sch

STARTER SKELETON: no CM5 connections yet. Do not fabricate.
Root sheet
StudyVerse handheld calculator

Sheet: /
File: cm5_calculator.kicad_sch

Title: CM5 Calculator Carrier

Size: A4	Date: 2026-09-30	**Rev: 0.1-skeleton**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/7

Block 1: CM5 mating connectors

- Two 100-pin board-to-board connectors (J_GPIO pins 1-100, J_HSS pins 101-200)
- Mating part and stack height: UNVERIFIED. Secondary source lists Amphenol 10164227-1001A1RLF; confirm part, stack height and orientation against RP-008180-DS.
- Unverified pin table for cross-check only: ../research/cm5_connector_pins_UNVERIFIED.csv
- Wireless variant: antenna keep-out at board edge (dimensions TBD from datasheet)
- M2.5 mounting holes (positions TBD from datasheet)

Source rule: every CM5 connection on this sheet must cite RP-008180-DS (CM5 datasheet), RP-008182-DS / CM5IO KiCad files, or the Pi 5 schematics. None have been read yet.

STARTER SKELETON: no CM5 connections yet. Do not fabricate.

CM5 handheld calculator carrier

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Sheet: /CM5 Connectors/

File: cm5_connectors.kicad_sch

Title: CM5 Connectors

Size: A4	Date: 2026-09-30	**Rev: 0.1-skeleton**
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Block 2 + 3: Power path, battery protection, fuel gauge

- USB-C input: CC pull-downs, ESD on CC/D+/D-/VBUS
- 1S Li-ion charger with power-path management
- 5V boost for CM5 +5V input, sized from datasheet worst-case current (TBD, Stage 2)
- Battery protection: OVP / UVP / OCP / short circuit
- I2C fuel gauge -> CM5 I2C

OPEN: battery capacity and dimensions (not given yet)

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STARTER SKELETON: no CM5 connections yet. Do not fabricate.
CM5 handheld calculator carrier
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Sheet: /Power/
File: power.kicad_sch

Title: Power

Size: A4	Date: 2026-09-30	**Rev: 0.1-skeleton**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 3/7

Block 4: DSI display connector

- Target: Raspberry Pi Touch Display 2 (DSI + I2C touch)
- Connector type/pinout must match the official cable: UNVERIFIED
- 100 ohm differential pairs, length matched (rules from datasheet)

OPEN: Touch Display 2 vs smaller bare DSI panel

Source rule: every CM5 connection on this sheet must cite RP-008180-DS (CM5 datasheet), RP-008182-DS / CM5IO KiCad files, or the Pi 5 schematics. None have been read yet.

STARTER SKELETON: no CM5 connections yet. Do not fabricate.

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Sheet: /Display DSI/

File: display.kicad_sch

Title: Display DSI

Size: A4	Date: 2026-09-30	**Rev: 0.1-skeleton**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 4/7

Block 5: CSI camera connector

- Camera Module 3 compatible, top edge of board
- Connector pinout: UNVERIFIED (take from CM5IO files)
- CAM_GPIO / I2C routing per CM5IO reference

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STARTER SKELETON: no CM5 connections yet. Do not fabricate.
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Sheet: /Camera CSI/
File: camera.kicad_sch

Title: Camera CSI

Size: A4	Date: 2026-09-30	**Rev: 0.1-skeleton**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 5/7

Block 6: Keypad

- Proposed (to be justified in Stage 2): I2C keypad scanner IC with interrupt line to CM5, mainline Linux driver, no firmware. Alternative: small MCU as USB HID keyboard.

OPEN: key count and switch type

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Sheet: /Keypad/
File: keypad.kicad_sch

Title: Keypad

Size: A4	Date: 2026-09-30	**Rev: 0.1-skeleton**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 6/7

Block 7: USB-C data, power button, LEDs, boot control

- USB 2.0 data from USB-C to CM5 (for flashing / rpiboot)
- nRPIBOOT switch or pad for USB boot mode
- PWR_BUT momentary button
- LED_nPWR / LED_nACT status LEDs
- PMIC_ENABLE, EEPROM_nWP handling per datasheet: UNVERIFIED

Block 8 (optional, if space): speaker/haptic footprint, expansion header

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Sheet: /USB Boot Control/

File: usb_boot.kicad_sch

Title: USB Boot Control

Size: A4	Date: 2026-09-30	**Rev: 0.1-skeleton**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 7/7